NAME : SAIF RAHMAN SHAIK
INTERNSHIP : DevOps
WEEKLY TASK : 6
DATE : 31/08/2026

Infrastructure as Code (IaC) & Configuration Management

1. Infrastructure as Code (IaC)

Infrastructure as Code is the practice of defining and managing infrastructure using machine-readable configuration files instead of creating resources manually. IaC makes infrastructure repeatable, consistent, version-controlled, and easier to automate.

Benefits: automation, consistency, version control, repeatability, faster provisioning, and reduced manual errors.

2. Terraform

Terraform is an open-source Infrastructure as Code tool used to provision and manage infrastructure through declarative configuration files. It can manage cloud and other infrastructure resources through providers.

Typical workflow: Write configuration → terraform init → terraform plan → terraform apply → verify → terraform destroy when required.

4. Terraform State

Terraform state records information about the infrastructure managed by Terraform. The state allows Terraform to compare the desired configuration with the existing resources and determine what needs to be created, changed, or removed. In real environments, teams commonly use a secure remote backend for shared state.

5. Variables & Outputs

Variables make Terraform configurations reusable by allowing values such as environment, region, VM size, or resource names to be supplied without changing the main configuration. **Outputs** display useful information after Terraform creates resources, such as an IP address, resource ID, or file path.

6. Terraform Modules

A Terraform module is a reusable collection of Terraform configuration files. Modules help standardize infrastructure and avoid repeating the same configuration. For example, an organization can create a reusable network or virtual-machine module and use it across multiple environments.

7. Ansible

Ansible is an automation and configuration-management tool used to configure servers, deploy applications, install packages, manage users, and control services. Ansible is generally agentless and commonly connects to Linux servers using SSH.

8. Ansible Inventory

An inventory identifies the hosts that Ansible manages. Hosts can be grouped according to their role, such as webservers, databases, or application servers. Inventory information can contain hostnames, IP addresses, groups, and connection variables.

9. Ansible Playbooks

An Ansible playbook is a YAML file containing plays and tasks that describe how systems should be configured. Playbooks make server configuration repeatable and can execute multiple tasks in a defined order.

10. Terraform vs Ansible

Terraform	Ansible
Primarily infrastructure provisioning	Primarily configuration management and automation
Creates and manages resources	Configures existing servers and applications
Uses .tf configuration files	Uses YAML playbooks
Uses providers	Uses modules
Maintains Terraform state	Does not use Terraform-style state for inventory/configuration
Example: create an Azure VM	Example: install Nginx on the VM